

Neža Žager Korenjak, PhD

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Research interests: hyperbolic & affine geometry, (higher) Teichmüller theory, geometric group theory

Positions	University of Michigan postdoc under supervision of Dick Canary	<i>Fall 2024-current</i>
	Max Plank Institute Leipzig postdoc under supervision of Anna Wienhard	<i>Fall 2023 - Summer 2024</i>
Education	University of Texas at Austin Ph.D. Program in Mathematics advisor: Jeffrey Danciger Ph.D. thesis: <i>Constructing proper affine actions via higher strip deformations.</i>	<i>Spring 2023</i>
	University of Ljubljana Bologna second cycle master's study program, Mathematics	<i>2016</i>
	Master's thesis: <i>Hypercyclic Operators</i> ◊ advisor: Roman Drnovšek	
	Bologna first cycle academic study program, Mathematics	<i>2014</i>
	Undergraduate thesis: <i>Theorems on Compactness</i> ◊ advisor: Žiga Virk	
Papers	<i>Proper affine deformations of positive representations.</i> Joint with J.-P. Burelle. arXiv: 2405.14658 (submitted)	<i>2024</i>
	<i>Constructing proper affine actions via higher strip deformations.</i> arXiv: 2205.14712 (submitted)	<i>2022</i>
	<i>Theorems on compactness.</i> Overview article. <i>Matrika</i> 1 (2014) 4 .	
Selected research talks & presentations	<i>Proper affine deformations of positive representations</i>	<i>May 2025</i>
	Higher rank geometric structures, Higgs bundles and physics, IHP Paris	
	<i>Constructing proper affine actions</i>	<i>April 2025</i>
	Lorentzian, Affine, and Hyperbolic Differential Geometry – in Memory of Todd Drumm, University of Maryland	
	<i>Proper affine deformations of positive representations</i>	<i>October 2024</i>
	2024 Fall Western Sectional Meeting, UC Riverside, Special Session on Geometry, Topology and Dynamics of Character Varieties.	
	<i>Proper affine deformations of positive representations</i>	<i>July 2024</i>
	AMS-UMI International Joint Meeting, Palermo, Anosov Representations and Higher Teichmüller Theory Special Session	
	<i>Proper affine deformations of positive representations</i>	<i>April 2024</i>
	University of Texas in Austin geometric geometry seminar	
	<i>Constructing proper affine actions via higher strip deformations</i>	<i>September 2023</i>
	Diverse Aspects of Groups, Geometry and Dynamics, Heidelberg University.	
	<i>Constructing proper affine actions via higher strip deformations</i>	<i>December 2022</i>
	University of Maryland geometry seminar	
	<i>Constructing proper affine actions via higher strip deformations</i>	<i>September 2022</i>
	University of Virginia geometry seminar	
	<i>Constructing proper affine actions via higher strip deformations</i>	<i>September 2022</i>
	Nearly carbon neutral topology & geometry conference (virtual.)	
	<i>Constructing proper affine actions</i>	<i>July 2022</i>
	GT GAPS (virtual.)	
	<i>Proper affine actions of free groups</i> , lightning talk.	<i>June 2022</i>
	Geometry, Arithmetic, and Groups, UT Austin	

	<i>Constructing proper affine actions</i> TWIMS 2022, Rice	February 2022
	<i>Constructing proper affine actions</i> The University of Sherbrooke, SAG seminar	October 2021
Teaching	Instructor at University of Michigan <i>Principles of real analysis (351)</i> <i>Explorations in Euclidean geometry (431)</i>	Fall 2024, Winter & Fall 2025 Fall 2025
	Assistant Instructor at the University of Texas at Austin <i>Learning assistant experience in math (M 371E)</i> A class for undergraduates about math pedagogy, where they get hands-on experience tutoring at CalcLab, which I am also organizing.	Fall 2021 - Spring 2023
	Summer mini-course leader <i>Measured laminations and train tracks</i> Led a weeklong online course aimed at graduate students as part of UT's Summer minicourse program	Summer 2021
	Mentor at the University of Texas at Austin's Geometry Lab Mentored a small group of undergraduates in a computer visualization problem in projective geometry, helped make the project's website	Spring 2021
	Teaching Assistant at the University of Texas at Austin <i>Differential Equations with Linear Algebra (M 427J)</i> <i>Honors Advanced Calculus for Applications II (M 427L)</i> <i>Advanced Calculus for Applications II (M 427L)</i> <i>Differential Calculus (M 408K)</i>	Spring 2020 Fall 2019 Spring 2019 Fall 2018
	Teaching Assistant at the University of Ljubljana <i>Computer Practical</i> <i>Mathematics 1, Grader</i>	Fall 2015 Spring 2015 - Summer 2016
	Additional training and certificates <i>Inclusive Classrooms Leadership Certificate</i> <i>Inclusive Instructor Certificate</i>	UT Austin, 2022 UT Austin, 2022
Outreach	Instructor at Ypsilanti MathCorps 2025 Taught a class of 8th grade math and a class of high school precalculus in a Summer program for kids from local underserved middle and high schools	Summer 2025
	DRP Mentor at the University of Texas at Austin Mentored undergraduates in reading projects on hyperbolic geometry, knot theory, low-dimensional topology, number theory, differential geometry	Spring 2017 - Spring 2023
	Volunteer at a tutoring pod at local middle school Helped middle school math teachers hold virtual office hours for students.	Fall 2020
	Speaker and volunteer at SMMG <i>an outreach program about mathematics aimed at middle and high schoolers</i> Regular volunteer & gave two talks: <i>An invitation to non-Euclidean geometry</i> <i>The Towers of Hanoi</i>	Spring 2020 Fall 2018
	UT Math Club speaker <i>Hyperbolic geometry</i>	Spring 2020
	Co-organizer and mentor of MaRS <i>a Slovenian summer math camp for high schoolers</i>	August 2015
	Volunteer tutor for undergraduate students at University of Ljubljana	Fall 2013- Spring 2016
	Leader and speaker of a math club for middle schoolers <i>dealing with fun math problems and preparations for math competitions</i>	2010-2011

Service	Co-organizer of a one-day conference <i>Queer in Math day</i>	<i>June 2024, MPI Leipzig</i>
	A one-day conference highlighting the work of queer mathematicians and promoting equity and diversity in math	
	Graduate Representative	
	serving on an elected committee of math grad students that helps deal with grad student concerns and well-being, organizes social and academic events, and helps mediate between students and faculty	<i>2020-2021, 2022-2023</i>
	Graduate Student Association	<i>2020-2021</i>
	Represented the math department graduate students in the legislative association of all UT Austin grad students	
	Organizer of the <i>Junior Topology Seminar</i>	<i>Spring 2020</i>
	The University of Texas at Austin	
	Co-organizer of the <i>Geometric Topology seminar</i>	<i>Spring 2018-Spring 2022</i>
	The University of Texas at Austin	
Honors and Awards	Co-organizer of the <i>Teichmüller theory learning seminar</i>	<i>Spring 2019</i>
	The University of Texas at Austin	
	Co-organizer of Directed Reading Program , UT Austin	<i>Spring 2019 - Spring 2021</i>
	<i>pairing graduate students in mathematics and related fields with undergraduate students interested in getting started in mathematical research to work on a reading project together, and organising the symposium for students to present their work</i>	
	Frank Gerth III teaching excellence award $\diamond$ <i>UT Austin</i>	<i>2020</i>
	Provost's Graduate Excellence Fellowship $\diamond$ <i>UT Austin</i>	<i>2016 - 2021</i>
	State merit scholarship (Zois scholarship)	<i>2007 - 2014</i>
	<i>rewarded by Republic of Slovenia</i>	
	Dean's Award for academic achievement in the first year of study	<i>2012</i>
	<i>University of Ljubljana</i>	
	Honorable mention at International Mathematical Olympiad	<i>2011</i>
	Bronze medal at Middle-European Mathematical Olympiad	<i>2010</i>